



EG1113 - C Programming for Engineering Lab

Report #2

For Loops, Scanf, and Bash Commands

Date of the experiment: August 28, 2024

Due date of report: August 30, 2021

Name: Nicholas Galus

1. Core concepts learned

- Setup new directories
- Utilize for loops to output print statements
- Utilized bash command line exit codes

2. List of equipment and software used

- Raspberry PI 400
- Notepad++
- Lenovo Laptop
- Caja File Network Share
- Visual Studio Code
- Mata Terminal

3. Relevant web links and other documents:

- Utilized [W3Schools](#) to create string variable types by utilizing character arrays
- Utilized [Red Hat](#) to use and understand bash command lines in regards to exit codes

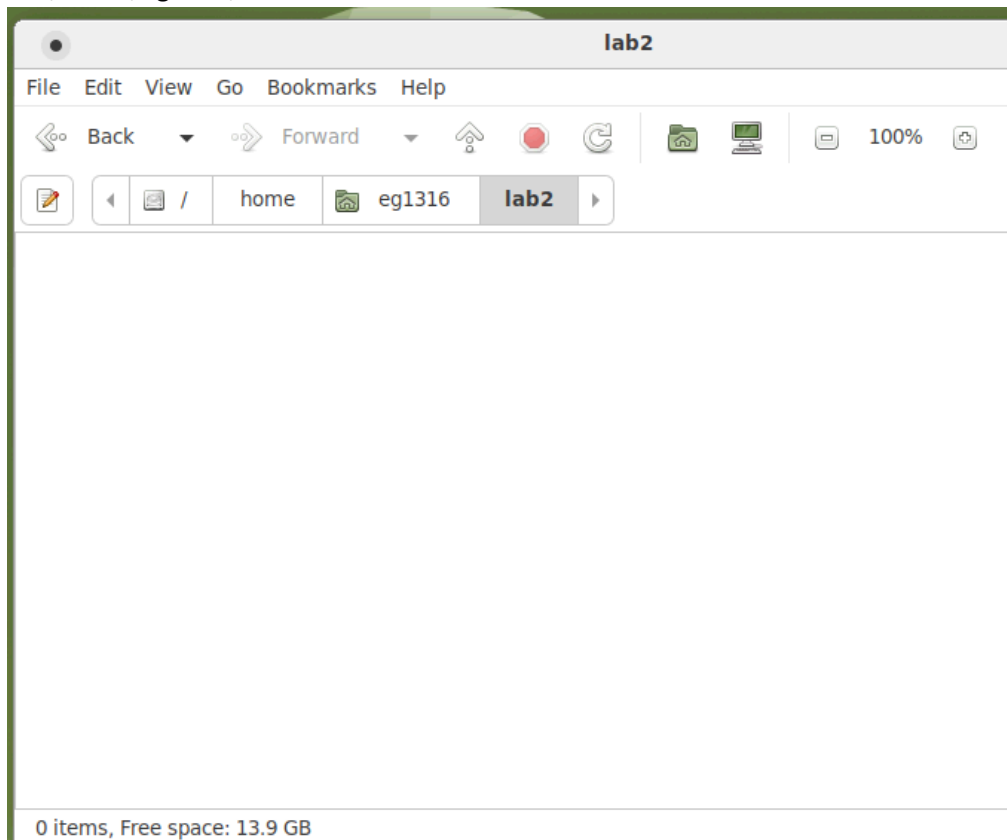
4. Exercise 1: Create a program with your initials

4.1 Summary

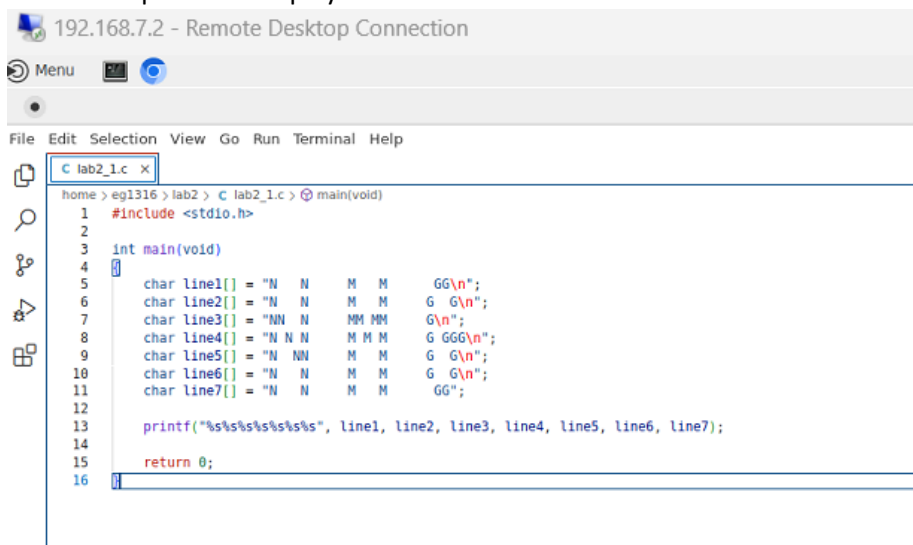
- Created a new directory for lab two with the directory: /home/eg1316/lab2
- Created a program named “lab2_1.c” to display your initials as a 5x7 matrix

4.2 Procedures

1. Create a new program named “lab2_1” stored in a new folder named “lab2” with the directory of “/home/eg1316/lab2”



2. Utilize one printf to display the initials “NMG” in a 5x7 matrix characters



3. Compile and run “lab2_1.c”

```

eg1316@eg1316-desktop:~/lab2
File Edit View Search Terminal Help
eg1316@eg1316-desktop:~$ cd lab2
eg1316@eg1316-desktop:~/lab2$ gcc -o lab2_1 lab2_1.c
eg1316@eg1316-desktop:~/lab2$ ./lab2_1
  N  N  M  M  GG
  N  N  M  M  G  G
 NN  N  MM MM  G
 N  N  N  M  M  M  G  GG
 N  NN  M  M  G  G
  N  N  M  M  G  G
  N  N  M  M  GG
eg1316@eg1316-desktop:~/lab2$

```

Free space: 13.8 GB

5. Exercise 2: Modify program from Exercise 1

5.1 Summary

- Modify the program from Exercise 1 to print the initials “NMG” 100 times

5.2 Procedures

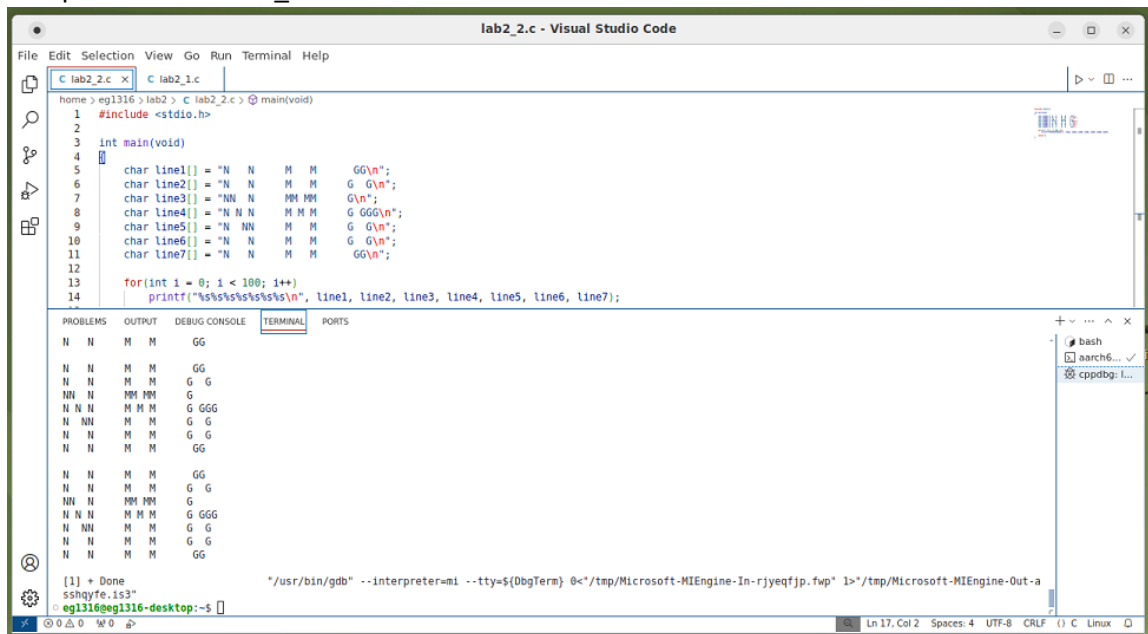
1. Duplicate “lab2_1” and rename to “lab2_2”
2. Add a for loop to print the initials “NMG” 100 times

```

home > eg1316 > lab2 > C lab2_2.c > main(void)
1  #include <stdio.h>
2
3  int main(void)
4  {
5      char line1[] = "N  N  M  M  GG\n";
6      char line2[] = "N  N  M  M  G  G\n";
7      char line3[] = "NN  N  MM MM  G\n";
8      char line4[] = "N  N  N  M  M  M  G  GG\n";
9      char line5[] = "N  NN  M  M  G  G\n";
10     char line6[] = "N  N  M  M  G  G\n";
11     char line7[] = "N  N  M  M  GG\n";
12
13     for(int i = 0; i < 100; i++)
14         printf("%s%s%s%s%s\n", line1, line2, line3, line4, line5, line6, line7);
15
16     return 0;
17

```

3. Compile and run "lab2_2.c"



```

lab2_2.c - Visual Studio Code
File Edit Selection View Go Run Terminal Help
C lab2_2.c x C lab2_1.c
1 home > eg1316 > lab2 > c lab2_2.c > main(void)
2 #include <stdio.h>
3
4 int main(void)
5 {
6     char line1[] = "N N M M GG\n";
7     char line2[] = "N N M M G G\n";
8     char line3[] = "NN N MM MM G\n";
9     char line4[] = "N N N M M M G GG\n";
10    char line5[] = "N NN M M G G\n";
11    char line6[] = "N N M M GG\n";
12
13    for(int i = 0; i < 100; i++)
14        printf("%s%s%s%s\n", line1, line2, line3, line4, line5, line6, line7);
15
16    return 0;
17 }
18
19 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
20 N N M M GG
21 N N M M G G
22 NN N MM MM G
23 N N N M M M G GG
24 N NN M M G G
25 N N M M G G
26 N N M M GG
27
28 N N M M GG
29 N N M M G G
30 NN N MM MM G
31 N N N M M M G GG
32 N NN M M G G
33 N N M M GG
34
35 [1] + Done
36 sshyfe.133"
37 eg1316eg1316-desktop:~$
38
39 "/usr/bin/gdb" --interpreter=mi --tty=${DbgTerm} 0<"/tmp/Microsoft-MIEngine-In-rjyeqfjp.fwp" 1>"/tmp/Microsoft-MIEngine-Out-a
40
41 Ln 17, Col 2 Spaces: 4 UTF-8 CRLF (1) C Linux

```

6. Exercise 3: Modify program from Exercise 2

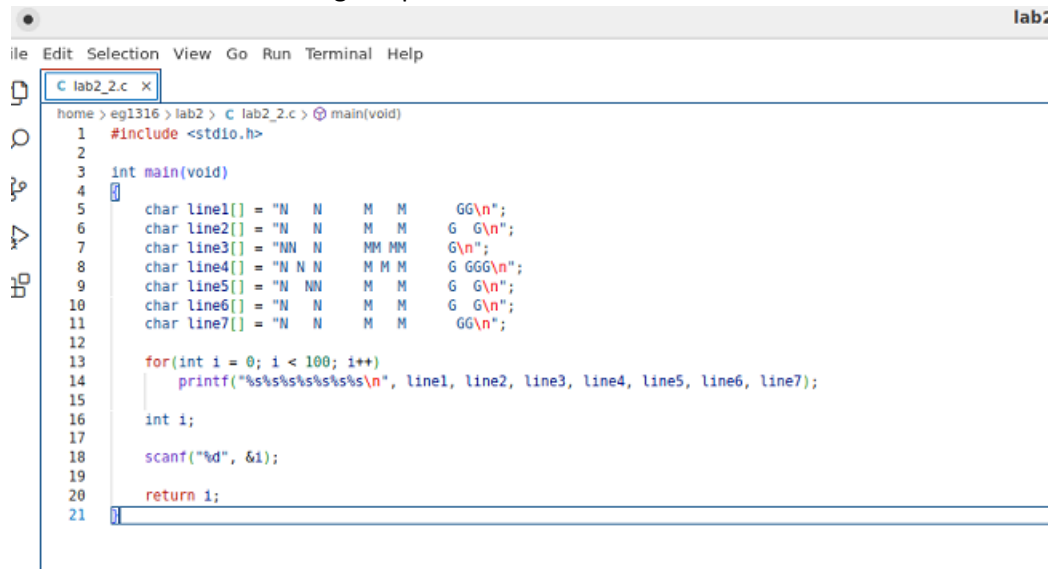
6.1 Summary

- Modify the program from Exercise 2 to accept an integer to be utilized as a return value
- Utilized bash command line exit codes
- Lab verification connecting to Wi-Fi at 2:20 pm



6.2 Procedures

1. Add scanf to collect an integer input and to return it

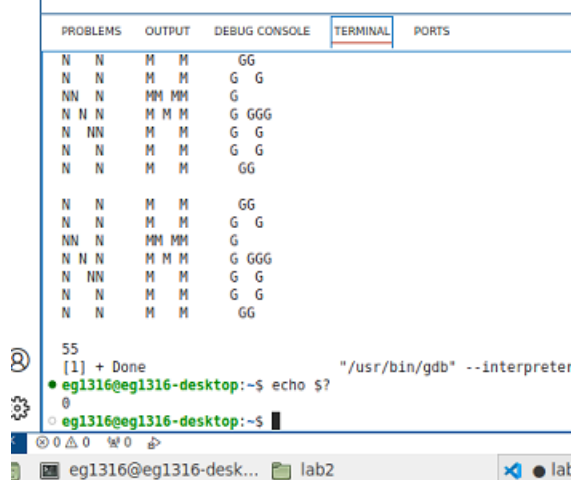


```

lab:
file Edit Selection View Go Run Terminal Help
C lab2_2.c x
home > eg1316 > lab2 > C lab2_2.c > main(void)
1  #include <stdio.h>
2
3  int main(void)
4  {
5      char line1[] = "N  N  M  M  GG\n";
6      char line2[] = "N  N  M  M  G  G\n";
7      char line3[] = "NN N  MM MM  G\n";
8      char line4[] = "N N N  M M M  G GGG\n";
9      char line5[] = "N NN  M  M  G  G\n";
10     char line6[] = "N  N  M  M  G  G\n";
11     char line7[] = "N  N  M  M  GG\n";
12
13     for(int i = 0; i < 100; i++)
14         printf("%s%s%s%s%s\n", line1, line2, line3, line4, line5, line6, line7);
15
16     int i;
17
18     scanf("%d", &i);
19
20     return i;
21 }

```

2. Compile and run program inputting 55, then checking for the previous return code by utilizing the bash command "echo \$?"



PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS
N N M M GG				
N N M M G G				
NN N MM MM G				
N N N M M M G GGG				
N NN M M G G				
N N M M G G				
N N M M GG				
N N M M GG				
N N M M G G				
NN N MM MM G				
N N N M M M G GGG				
N NN M M G G				
N N M M G G				
N N M M GG				

```

55
[1] + Done "/usr/bin/gdb" --interpreter
eg1316@eg1316-desktop:~$ echo $?
0
eg1316@eg1316-desktop:~$

```

7. Problems experienced and areas of concern

- Obtained different outputs when running the "echo \$?" command in the Mata Terminal and Visual Studio Terminal

8. Course goals

Rank the level (0=none to 5=a large amount) that this lab covered the each of the course goals by updating the appropriate box in the table below.

[illegible]